Online Dynamic Resource Allocation with State Constraints

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Abstract

We study online dynamic resource allocation with resource-in-storage availability constraints under adversarial inflows and convex costs through the lens of adversarial online control. At each round, the controller chooses an allocation within available resources before observing the inflow and cost. We introduce *Simplex Disturbance-Action Control* (SDAC), a simplex-constrained disturbance-action policy class that preserves feasibility by construction, and we develop an online controller over SDAC weights. We benchmark this algorithm against the best fixed SDAC policy in hindsight. Our regret analysis yields sublinear dependence on the horizon and logarithmic dependence on the policy dimension. The setup is motivated by applications in energy-harvesting battery management where decisions must satisfy storage-availability limits in real time.

1 Introduction

1.1 Motivation and Problem Setting

Many dynamic resource systems are governed by a budget state variable that evolves over time and constrains the allocation action at every round. The controller must choose an allocation action from currently available resources before observing the round's inflow and cost.

This creates an online optimization problem with state coupling: aggressive early allocation can harm later performance, while overly conservative allocation can increase immediate cost. Our objective is to design a policy that remains feasible at every round and competes with the best fixed policy in a structurally tractable comparator class.

1.2 Model and Information Structure

We study a discrete-time storage-allocation system with state $b_t \geq 0$, where b_t represents the currently available resource. At each time t, the controller chooses an allocation action u_t under the *feasibility constraints*:

$$0 \leq u_t \leq b_t,$$

which means the controller cannot allocate more than the currently available resource or any negative amount. After the action is chosen, an adversarial inflow a_t arrives, and the state evolves according to

$$b_{t+1} = b_t - u_t + a_t,$$

with inflow a_t.

Unlike stochastic application models that rely on distributional assumptions, our formulation is adversarial. Inflows a_t and convex costs $c_t(b, u)$ may vary arbitrarily over time, and the controller chooses u_t before observing either object at round t. The goal is low cumulative cost relative to the best fixed policy in a later-defined comparator class, while maintaining feasibility throughout.

1.3 Application Domains

This abstraction captures a broad set of domains:

- **Energy Harvesting and Battery Management:** b_t is battery state of charge, a_t is harvested energy, and u_t is discharge or consumption.

- **Queueing Networks and Buffer Management:** b_t is queue/buffer occupancy, a_t is arrivals, and u_t is service rate constrained by current backlog.

- **Inventory Control:** b_t is on-hand inventory, a_t is supply or replenishment, and u_t is sales under stock constraints.

- **Investments and Bank Accounts:** b_t is liquid capital/balance, a_t is incoming cash flow, and u_t is capital deployment or withdrawal under budget limits.

- **Water Reservoir and Dam Management:** b_t is stored water, a_t is inflow, and u_t is controlled release under storage limits.

Queueing and Network Optimization An alternative, though mathematically related, perspective comes from queueing control in computer networks and telecommunications. In that framework, one tracks a queue with stochastic arrivals and makes real-time service decisions (equivalent to allocations in our setup), typically aiming to minimize long-run average service cost while keeping queue lengths stable. This literature is extensive (see, e.g., [16, 17, 11, 13]), but its setting differs from ours in several important respects. Classical queueing work usually assumes stochastic arrivals and focuses on simple, known cost functions that depend only on the service rate, while queue length is kept small. These methods achieve trade-offs between queue size and time-average service cost. By contrast, our model permits fully adversarial sequences of inflows and costs and allows general convex costs that depend on both the queue length (state) and the service rate (action). Thus our approach is more general in cost structure and input model. The queueing literature provides useful motivation and techniques for state-coupled dynamics, but does not directly address the online optimization and regret-minimization objectives we study.

Another related work is [8], which proposes and analyzes a fixed controller that is a linear function of past arrivals. That controller resembles our family of SDAC policies because it uses nonnegative filter weights that sum to one. However, the controller in [8] is not updated online and is chosen to achieve bounded service delay; by contrast, our framework focuses on online optimization of a general cost and does not treat delay as a constraint.

Inventory Management A related work is [18], which studies a distinct model in which demand (equivalent to allocation in our setup) is chosen adversarially, while the controller chooses the supply (equivalent to inflows in our model). The objective is to minimize time-average purchasing cost while always meeting demand, and the performance metric is competitive ratio rather than regret. Although structurally similar to our setting, the control variable differs: we choose allocation from available resources, whereas [18] chooses how much to procure. Another key difference is the loss model: our framework allows general convex state-action costs, while [18] uses a linear cost depending only on the action.

More closely related to ours is [7], which considers an inventory-control formulation with potentially nonlinear and nondifferentiable state-transition functions, and studies base-stock policies. While that model is more general in scope, its decision structure differs from ours: as in [18], the controller chooses supply (inflow) and then observes demand (allocation) through the transition, whereas our controller chooses allocation and observes incoming supply inflows. Methodologically, our setup is intentionally narrower and tailored to SDAC-based adversarial regret analysis: feasibility constraints are enforced by policy design, yielding a simpler linear state recursion, lower computational overhead, and cleaner analysis than approaches operating through nonlinear nondifferentiable dynamics such as [7]. The two frameworks are structurally related but target different problem structures, control variables, and policy classes.

Energy Harvesting and Battery Management A closely related work is [10], which studies an online energy-harvesting allocation problem with dynamics similar to ours. That work benchmarks against the best sequence of decisions in hindsight, but it differs in several important ways: [10] uses competitive ratio rather than regret, adopts an action-only cost model rather than general convex state-action costs, which means in particular that it cannot optimize a function of the battery size, and it has a less adversarial information pattern because the algorithm observes the current cost before selecting the action, whereas in our model the action is chosen before the cost is revealed.

Prior work on energy harvesting includes [3], which improves over [19] by achieving $O(\sqrt{T})$ regret with finite battery and adversarial costs (versus the stochastic costs in [19]). Both works consider action-only costs (not state-dependent), i.i.d. inflows (not adversarial), and benchmark against fixed allocation amounts rather than fixed policies. In contrast, our framework jointly optimizes over state-action costs, fully adversarial sequences, and stronger comparators, requiring new algorithmic techniques.

Water Reservoir and Dam Management Classical dam theory studies storage dynamics with random inflows and controlled releases (see, for example, [12, 14]). Relative to our adversarial online-control setting, this literature typically assumes stochastic inputs and emphasizes stationary performance criteria under simple, known cost structures. Here, it serves mainly as historical context for state-coupled allocation under feasibility constraints.

Online control A recent line of work in online learning studies online control, where a linear dynamical system is operated under adversarial disturbances and costs. The objective is low time-average cost relative to the best policy in hindsight, rather than classical stabilization or worst-case guarantees. Foundational results for linear systems and adversarial settings include [1, 2, 5, 15]; see [6] for a modern overview.

A central idea in this literature is *Disturbance-Action Control* (DAC), a policy parameterization in which control actions are expressed as affine or linear functions of the current state and a finite history of past disturbances. This representation yields tractable surrogates for policy optimization and enables efficient no-regret updates in adversarial environments. Extensions include partial observability [9] and bandit feedback [5]. SDAC adapts this template to state-constrained resource systems, where feasibility is coupled over time through the dynamics.

Research Gap and Paper Scope Online control provides a strong foundation for our problem class: it accommodates arbitrary time-varying inflows (disturbances in control language) and unknown general convex costs, and the resulting algorithms come with robust guarantees and tractability. However, for budget-coupled allocation dynamics, vanilla DAC does not enforce per-round feasibility by construction. Our framework fills this gap by replacing DAC with SDAC, a simplex-constrained DAC-style family that preserves the feasibility defined by (1) while retaining the desirable characteristics.

1.4 Contributions

The contributions are organized in two layers:

- **Modeling and policy-class contribution.** We formulate adversarial online dynamic resource allocation with state-coupled feasibility and state-dependent costs, where the cost depends on the stored amount in the storage, and we introduce *Simplex Disturbance-Action Control* (SDAC), a DAC-inspired simplex-constrained policy class to compete with. The SDAC *enforces feasibility* by construction while preserving DAC-style expressiveness.

- **Algorithmic and analytical contribution.** We develop an exponentiated-gradient online controller over SDAC weights in which a clipping step *enforces feasibility* of the online action, and we analyze regret against the best fixed SDAC comparator in hindsight, achieving sublinear regret with logarithmic dependence on policy dimension.

1.5 Paper Organization

Section 3 formalizes the adversarial allocation model, assumptions, and regret benchmark. Section 4 introduces the SDAC policy class and summarizes its structural properties. Section 5 presents the online controller, Section 6 develops regret guarantees, and Section 7 reports experiments.

2 Notation

We denote by $t \in \{1, 2, \ldots, T\}$ the time index (total horizon T). The system state $b_t \in \mathbb{R}_{\geq 0}$ is the available resource, and the action $u_t \in \mathbb{R}_{\geq 0}$ satisfies $0 \leq u_t \leq b_t$. The exogenous inflow is $a_t \in [0, 1]$, and the cost function is $c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$. The online algorithm is $\mathcal{A}$, and π (or $\pi(w)$ when parameterized by w) denotes a policy from the comparator class $\Pi = \mathrm{SDAC}^{\kappa}_{H}$. The parameters $\kappa \in (0, 1]$ (gain) and $H \in \mathbb{N}$ (memory horizon) are fixed design parameters. The weight vector w lies in the unit simplex $\Delta_H = \{w \in \mathbb{R}^H_{\geq 0} : \sum_i w^{(i)} \leq 1\}$. The Lipschitz constants for the cost function are $L_b, L_u > 0$. The inner product is denoted $\langle x, y \rangle = x^T y$.

Throughout the paper, several terms are used interchangeably for the same objects. The state variable b_t may also be described as the budget, storage level, battery state, or inventory level, depending on the application. The action u_t may also be called the allocation, outflow, release, discharge, or consumption decision. Likewise, the inflow a_t may appear as arrival, supply, harvested energy, or replenishment. The constraint $0 \leq u_t \leq b_t$ is the feasibility, availability, or state-constraint condition, and the cost $c_t(b_t, u_t)$ is referred to as the loss, or stage cost.

3 Model and Problem Setup

We consider a discrete-time allocation process over rounds $t \in \{1, 2, \ldots, T\}$. The system state is denoted by $b_t \in \mathbb{R}_{\geq 0}$, where b_t represents the available budget at the beginning of round t. The initial state is given as

$$b_1 = 0.$$

At each round, the controller chooses an allocation action $u_t \in \mathbb{R}_{\geq 0}$ subject to state feasibility:

$$0 \leq u_t \leq b_t. \tag{1}$$

The system then receives an exogenous inflow $a_t \in [0, 1]$ and evolves according to

$$b_{t+1} = b_t - u_t + a_t. \tag{2}$$

After the action is selected, a convex cost function

$$c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$$

is revealed, and the incurred cost is $c_t(b_t, u_t)$. We assume each c_t is convex in (b, u) and satisfies a Lipschitz condition on the relevant domain. There exist constants $L_b, L_u > 0$ such that for all admissible (b, u) and (b', u'),

$$|c_t(b, u) - c_t(b', u')| \leq L_b|b - b'| + L_u|u - u'|. \tag{3}$$

The information pattern is adversarial and non-anticipatory. At time t, the controller chooses u_t before observing a_t and c_t. Hence both the inflow sequence $\{a_t\}_{t=1}^T$ and the cost sequence $\{c_t\}_{t=1}^T$ may be arbitrary.

Assumptions Equations (1)–(3) define the model. For analysis, we collect the remaining standing assumptions as follows:

1. **Inflow bounds:** $a_t \in [0, 1]$ for all t.
2. **Adversarial timing:** at round t, u_t is chosen before observing a_t and c_t.
3. **Cost regularity:** each c_t is convex in (b, u) and obeys (3).
4. **Feasible policy class:** both the online policy and comparator policies are required to satisfy (1) under dynamics (2).

For an online algorithm $\mathcal{A}$ producing trajectory $\{(b_t^{\mathcal{A}}, u_t^{\mathcal{A}})\}_{t=1}^T$, define cumulative cost

$$J_T(\mathcal{A}) \triangleq \sum_{t=1}^{T} c_t\big(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big). \tag{4}$$

For fixed (κ, H), let the comparator class be $\Pi \triangleq \mathrm{SDAC}_H^\kappa$ (defined in Section 4). For each $\pi \in \Pi$, let $\{(b_t^\pi, u_t^\pi)\}_{t=1}^T$ be the induced feasible trajectory under the same inflow sequence. Regret is defined by

$$\mathrm{Regret}(T) \triangleq \sum_{t=1}^{T} c_t\big(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - \min_{\pi \in \Pi} \sum_{t=1}^{T} c_t\big(b_t^\pi, u_t^\pi\big). \tag{5}$$

While the online learner may update its SDAC weights over time, the benchmark in (5) is the best single fixed SDAC policy chosen in hindsight. The objective is to design an online policy that preserves feasibility in (1) for all t and achieves sublinear regret as defined in (5).

4 Simplex Disturbance-Action Control (SDAC)

4.1 From Disturbance-Action Control to SDAC

Disturbance-Action Control (DAC) parameterizes the action as a linear function of the current state and a finite window of past disturbances. This representation is central in adversarial online control because it accommodates arbitrary disturbance and cost sequences, yields robust regret guarantees, and enables efficient online convex-optimization updates. In our setting, a representative DAC form is

$$u_t = \kappa b_t + \sum_{i=1}^{H} m^{(i)} a_{t-i}, \tag{6}$$

where $\kappa \in \mathbb{R}$ is a gain, $H \in \mathbb{N}$ is the memory horizon, and $m \in \mathbb{R}^H$ is a disturbance filter.

For our constrained allocation model, raw DAC is insufficient: (6) does not generally enforce (1) at every round. In particular, the action can violate either bound, with $u_t > b_t$ or $u_t < 0$. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which replaces unconstrained DAC filters by a simplex-constrained family tailored to budget dynamics. SDAC is designed to keep the DAC advantages above (adversarial robustness, regret-based comparison, and tractable online updates) while adding feasibility by construction.

4.2 SDAC Definition

Fix $\kappa \in (0, 1]$ and memory horizon $H \in \mathbb{N}$. Let

$$\Delta_H \triangleq \left\{ w \in \mathbb{R}^H_{\geq 0} : \sum_{i=1}^{H} w^{(i)} \leq 1 \right\} \tag{7}$$

be the unit simplex. For any $w \in \Delta_H$, define the policy $\pi(w)$ by

$$u_t^{\pi(w)} \triangleq \kappa b_t^{\pi(w)} + \sum_{i=1}^{H} w^{(i)} (1-\kappa)^i a_{t-i}, \qquad a_\tau \equiv 0 \text{ for } \tau \leq 0. \tag{8}$$

For each round t, let

$$\widetilde{a}_t \triangleq \big((1-\kappa)^1 a_{t-1}, \ldots, (1-\kappa)^H a_{t-H}\big)^\top \in \mathbb{R}^H,$$

so that (8) can be written as

$$u_t^{\pi(w)} = \kappa b_t^{\pi(w)} + \langle w, \widetilde{a}_t \rangle. \tag{9}$$

The induced state trajectory satisfies

$$b_{t+1}^{\pi(w)} = b_t^{\pi(w)} - u_t^{\pi(w)} + a_t, \qquad b_1^{\pi(w)} = 0. \tag{10}$$

The resulting class is

$$\mathrm{SDAC}_H^{\kappa} \triangleq \{\pi(w) : w \in \Delta_H\},$$

with Δ_H defined in (7).

First, feasibility is preserved within the class: for any fixed $w \in \Delta_H$, the induced trajectory under (8)–(10) satisfies $0 \le u_t^{\pi(w)} \le b_t^{\pi(w)}$ for all t (see Lemma 3 in Appendix A).

Second, SDAC is compatible with online convex optimization. For fixed inflows, both $b_t^{\pi(w)}$ and $u_t^{\pi(w)}$ are linear in w. Therefore, since each c_t is convex in (b, u), the induced cost is convex in w (see Lemma 4 in Appendix A). Regret minimization against the best fixed SDAC comparator thus becomes online convex optimization over the simplex.

5 Online Algorithm

We now define an online controller over SDAC weights using exponentiated-gradient updates on the simplex.

For each round t, let $w_t \in \Delta_H$ be the online weight vector.

Action selection The controller first forms the unconstrained SDAC-style action

$$\widehat{u}_t \triangleq \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle, \tag{11}$$

then enforces feasibility by clipping:

$$u_t^{\mathcal{A}} \triangleq \min\{\widehat{u}_t,\, b_t^{\mathcal{A}}\}. \tag{12}$$

The state update is

$$b_{t+1}^{\mathcal{A}} = b_t^{\mathcal{A}} - u_t^{\mathcal{A}} + a_t, \qquad b_1^{\mathcal{A}} = 0. \tag{13}$$

Feasibility of the online trajectory We verify that the algorithm satisfies the state-action constraints at every round. The key mechanism is the clipping step in (12), which enforces per-round feasibility by truncating the unconstrained proposal to the currently available state. Since $w_t \in \Delta_H$, $\widetilde{a}_t \geq 0$ componentwise, and $\kappa \in (0, 1]$, equation (11) gives $\widehat{u}_t \geq 0$. By (12),

$$u_t^{\mathcal{A}} = \min\{\widehat{u}_t, b_t^{\mathcal{A}}\} \in [0, b_t^{\mathcal{A}}],$$

so the action feasibility constraint holds whenever $b_t^{\mathcal{A}} \geq 0$. It remains to prove $b_t^{\mathcal{A}} \geq 0$ for all t. The base case is $b_1^{\mathcal{A}} = 0$. If $b_t^{\mathcal{A}} \geq 0$, then $u_t^{\mathcal{A}} \leq b_t^{\mathcal{A}}$ and $a_t \in [0, 1]$, so from (13),

$$b_{t+1}^{\mathcal{A}} = b_t^{\mathcal{A}} - u_t^{\mathcal{A}} + a_t \geq 0.$$

Thus, by induction, $b_t^{\mathcal{A}} \geq 0$ and $0 \leq u_t^{\mathcal{A}} \leq b_t^{\mathcal{A}}$ for all t.

Surrogate cost and gradient step Define the surrogate cost at round t by

$$\ell_t(w) \triangleq c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big), \qquad w \in \Delta_H, \tag{14}$$

which is convex and Lipschitz by Lemma 4. Let $g_t \in \partial \ell_t(w_t)$ be a subgradient.

The exponentiated-gradient update is

$$\widetilde{w}_{t+1}^{(i)} \triangleq w_t^{(i)} \exp\big(-\eta g_t^{(i)}\big), \qquad i = 1, \ldots, H, \tag{15}$$

followed by [1]:

$$w_{t+1}^{(i)} \triangleq \begin{cases} \dfrac{\widetilde{w}_{t+1}^{(i)}}{\sum_{j=1}^{H} \widetilde{w}_{t+1}^{(j)}}, & \text{if } \sum_{j=1}^{H} \widetilde{w}_{t+1}^{(j)} > 1, \\ \widetilde{w}_{t+1}^{(i)}, & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, H. \tag{16}$$

This yields a feasible online trajectory by (12) and performs online convex optimization over Δ_H through (14)–(16).

6 Regret Analysis

The regret proof has two parts: a standard online convex optimization bound for the surrogate losses ℓ_t, and a stability bound showing that the clipped online action stays close to the SDAC comparator trajectory driven by the same weights. The technical lemmas are deferred to Appendix A.

[1] This step is a (Bregman) projection onto the unit simplex normalization.

Theorem 1 (Regret bound)**.** *Suppose each cost function c_t is convex and satisfies* (3)*. Then the clipped exponentiated-gradient controller from Section 5 satisfies*

$$\mathrm{Regret}(T) \leq \frac{\log H}{\eta} + \eta L^2 T\Big(\tfrac{1}{2} + \tfrac{1}{\kappa^2}\Big), \qquad L := L_u + \tfrac{L_b}{\kappa}.$$

In particular, the choice

$$\eta^* = \sqrt{\frac{\log H}{L^2 T\big(\frac{1}{2} + \frac{1}{\kappa^2}\big)}}$$

gives

$$\mathrm{Regret}(T) \leq 2L\sqrt{\Big(\tfrac{1}{2} + \tfrac{1}{\kappa^2}\Big) T \log H}.$$

Proof sketch. By Lemma 4, the surrogate losses are convex and L-Lipschitz, so standard entropic OMD gives the learning term bound $\log H/\eta + \eta L^2 T/2$. The remaining stability term is controlled by Lemmas 5–11, which together contribute at most $\eta L^2 T/\kappa^2$. Combining the two bounds yields the claim. □

7 Experiments

A Appendix: Deferred Proofs

This appendix collects the deferred proofs for structural SDAC properties used in Section 4 and the regret bound stated in Section 6.

Recall that we set $a_\tau \equiv 0$ for $\tau \leq 0$, which is used in several base-case arguments below.

Lemma 1 (State decomposition under SDAC)**.** *Fix $\kappa \in (0, 1]$, $H \in \mathbb{N}$, and $w \in \Delta_H$. Let $\pi(w)$ be defined by* (8) *and* (10)*. Then for all $t \geq 2$,*

$$b_t^{\pi(w)} = \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i} (1-\kappa)^{j-1} a_{t-j} \right) = \langle w, \psi_t \rangle, \tag{17}$$

where

$$\psi_t^{(i)} \triangleq \sum_{j=1}^{i} (1-\kappa)^{j-1} a_{t-j}, \qquad i = 1, \dots, H.$$

In particular, $0 \leq \psi_t^{(i)} \leq 1/\kappa$. Consequently, since $w \in \Delta_H$ and $b_t^{\pi(w)} = \langle w, \psi_t \rangle$, we also have $b_t^{\pi(w)} \geq 0$ for all $t \geq 2$.

Proof. We use induction on t. For $t = 2$, $b_1^{\pi(w)} = 0$ and $a_{1-i} = 0$ for all $i \geq 1$. By the policy definition (8), $u_1^{\pi(w)} = \kappa b_1^{\pi(w)} + \sum_{i=1}^{H} w^{(i)}(1-\kappa)^i a_{1-i} = 0$. Then, using the state update (10), $b_2^{\pi(w)} = b_1^{\pi(w)} - u_1^{\pi(w)} + a_1 = a_1$, which is exactly (17) at time $t = 2$.

Assume the claim holds at time $t \geq 2$. Using (8) and (10),

$$b_{t+1}^{\pi(w)} = (1-\kappa) b_t^{\pi(w)} - \sum_{i=1}^{H} w^{(i)}(1-\kappa)^i a_{t-i} + a_t.$$

Substituting the induction hypothesis gives

$$\begin{aligned}
b_{t+1}^{\pi(w)} &= (1-\kappa) \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i} (1-\kappa)^{j-1} a_{t-j} \right) - \sum_{i=1}^{H} w^{(i)}(1-\kappa)^i a_{t-i} + a_t \\
&= \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i} (1-\kappa)^{j} a_{t-j} \right) - \sum_{i=1}^{H} w^{(i)}(1-\kappa)^i a_{t-i} + a_t \\
&= \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i-1} (1-\kappa)^{j} a_{t-j} \right) + a_t \\
&= \sum_{i=1}^{H} w^{(i)} \left(a_t + \sum_{j=1}^{i-1} (1-\kappa)^{j} a_{t-j} \right) \\
&= \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i} (1-\kappa)^{j-1} a_{t+1-j} \right),
\end{aligned}$$

which is exactly (17) at time $t+1$. In the third displayed line above we reindexed the inner summation (shifted the index $j \mapsto j+1$) to obtain the final form. The bound $\psi_t^{(i)} \leq 1/\kappa$ follows from $a_\tau \in [0,1]$ and the geometric series. □

Lemma 2 (Linearity of SDAC action in w)**.** *Under the setup of Lemma 1, for each t,*

$$u_t^{\pi(w)} = \langle w, \phi_t \rangle,$$

where

$$\phi_t^{(i)} \triangleq \kappa \psi_t^{(i)} + (1-\kappa)^i a_{t-i} = \sum_{j=1}^{i} (1-\kappa)^{j-1} a_{t-j}.$$

Moreover, $0 \leq \phi_t^{(i)} \leq 1$ *for all* i.

Proof. From (8),

$$u_t^{\pi(w)} = \kappa b_t^{\pi(w)} + \sum_{i=1}^{H} w^{(i)}(1-\kappa)^i a_{t-i}.$$

Applying Lemma 1, we substitute $b_t^{\pi(w)} = \langle w, \psi_t \rangle = \sum_{i=1}^{H} w^{(i)} \psi_t^{(i)}$, so

$$\begin{aligned} u_t^{\pi(w)} &= \kappa \sum_{i=1}^{H} w^{(i)} \psi_t^{(i)} + \sum_{i=1}^{H} w^{(i)}(1-\kappa)^i a_{t-i} \\ &= \sum_{i=1}^{H} w^{(i)} \big(\kappa \psi_t^{(i)} + (1-\kappa)^i a_{t-i}\big) \\ &= \sum_{i=1}^{H} w^{(i)} \phi_t^{(i)} = \langle w, \phi_t \rangle. \end{aligned}$$

The lower bound on $\phi_t^{(i)}$ is immediate from $a_{t-i} \geq 0$ and $\kappa \leq 1$. For the upper bound, use $a_\tau \in [0,1]$:

$$\phi_t^{(i)} \leq \kappa \sum_{j=1}^{i} (1-\kappa)^{j-1} + (1-\kappa)^i = 1.$$

□

Lemma 3 (Feasibility of SDAC policies)**.** *For every fixed $w \in \Delta_H$, the trajectory induced by* (8)–(10) *satisfies*

$$0 \leq u_t^{\pi(w)} \leq b_t^{\pi(w)} \quad \textit{for all } t \geq 1.$$

Proof. Nonnegativity of $u_t^{\pi(w)}$ follows from (8), because each term in the sum is nonnegative: $\kappa b_t^{\pi(w)} \geq 0$ since $\kappa > 0$ and $b_t^{\pi(w)} \geq 0$ (by Lemma 1 for $t \geq 2$, and $b_1^{\pi(w)} = 0$ by (10)), while $w^{(i)} \geq 0$, $(1-\kappa)^i \geq 0$, and $a_{t-i} \geq 0$ for every i.

For the upper bound, Lemma 1 implies

$$\begin{aligned} (1-\kappa) b_t^{\pi(w)} &= (1-\kappa) \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i} (1-\kappa)^{j-1} a_{t-j} \right) \\ &= \sum_{i=1}^{H} w^{(i)} \left(\sum_{j=1}^{i} (1-\kappa)^{j} a_{t-j} \right). \end{aligned}$$

Now rewrite the inner sum by collecting the coefficient of each a_{t-i}. Since all weights are nonnegative, for every i we have

$$\sum_{j=i}^{H} w^{(j)} \geq w^{(i)}.$$

Therefore,

$$(1-\kappa)b_t^{\pi(w)} = \sum_{i=1}^{H} \left(\sum_{j=i}^{H} w^{(j)} \right) (1-\kappa)^i a_{t-i} \geq \sum_{i=1}^{H} w^{(i)} (1-\kappa)^i a_{t-i},$$

since $\sum_{j=i}^{H} w^{(j)} \geq w^{(i)}$ and $a_{t-i} \geq 0$. Therefore,

$$u_t^{\pi(w)} = \kappa b_t^{\pi(w)} + \sum_{i=1}^{H} w^{(i)} (1-\kappa)^i a_{t-i} \leq \kappa b_t^{\pi(w)} + (1-\kappa) b_t^{\pi(w)} = b_t^{\pi(w)}.$$

□

Lemma 4 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(w) \triangleq c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big), \qquad w \in \Delta_H.$$

Then ℓ_t is convex on Δ_H, and for all $w, w' \in \Delta_H$,

$$|\ell_t(w) - \ell_t(w')| \leq \left(\frac{L_b}{\kappa} + L_u \right) \|w - w'\|_1.$$

Proof. By Lemma 1 and Lemma 2,

$$\ell_t(w) = c_t\big(\langle w, \psi_t \rangle, \langle w, \phi_t \rangle\big).$$

The map $w \mapsto (\langle w, \psi_t \rangle, \langle w, \phi_t \rangle)$ is affine, and c_t is convex, so ℓ_t is convex.

Using (3),

$$\begin{aligned} |\ell_t(w) - \ell_t(w')| &= \big| c_t\big(\langle w, \psi_t \rangle, \langle w, \phi_t \rangle\big) - c_t\big(\langle w', \psi_t \rangle, \langle w', \phi_t \rangle\big) \big| \\ &\leq L_b \big| \langle w - w', \psi_t \rangle \big| + L_u \big| \langle w - w', \phi_t \rangle \big| \\ &\leq L_b \|w - w'\|_1 \|\psi_t\|_\infty + L_u \|w - w'\|_1 \|\phi_t\|_\infty \\ &= \big(L_b \|\psi_t\|_\infty + L_u \|\phi_t\|_\infty \big) \|w - w'\|_1. \end{aligned}$$

Lemma 1 gives $\|\psi_t\|_\infty \leq 1/\kappa$, and Lemma 2 gives $\|\phi_t\|_\infty \leq 1$, which yields the claim. In particular, one may take the surrogate Lipschitz constant as

$$L := L_u + \frac{L_b}{\kappa},$$

as used in the main text. □

Lemma 5 (One-step movement of entropic OMD)**.** *For the entropic OMD update with step size $\eta > 0$ and subgradient $g_t \in \partial \ell_t(w_t)$,*

$$\|w_{t+1} - w_t\|_1 \le \eta \, \|g_t\|_\infty.$$

In particular, if $\|g_t\|_\infty \le L$, then $\|w_{t+1} - w_t\|_1 \le \eta L$.

Proof. Let $\Psi(w) = \sum_{i=1}^{H} w^{(i)} \log w^{(i)}$, and write $\Delta_t \triangleq w_{t+1} - w_t$. The entropic mirror-descent optimality condition gives, for all $w \in \Delta_H$,

$$\Big\langle g_t + \tfrac{1}{\eta}\big(\nabla\Psi(w_{t+1}) - \nabla\Psi(w_t)\big),\, w - w_{t+1} \Big\rangle \ge 0.$$

Taking $w = w_t$ and rearranging yields

$$\langle g_t, \Delta_t \rangle \le \tfrac{1}{\eta}\Big\langle \nabla\Psi(w_{t+1}) - \nabla\Psi(w_t),\, \Delta_t \Big\rangle.$$

By the Bregman identity,

$$\Big\langle \nabla\Psi(w_{t+1}) - \nabla\Psi(w_t),\, \Delta_t \Big\rangle = D_\Psi(w_{t+1}\|w_t) + D_\Psi(w_t\|w_{t+1}).$$

Hence, using Hölder's inequality,

$$D_\Psi(w_{t+1}\|w_t) + D_\Psi(w_t\|w_{t+1}) \le \eta \, \|g_t\|_\infty \, \|\Delta_t\|_1.$$

Since both iterates lie in the simplex, Pinsker's inequality (i.e., the bound $D_{KL}(p\|q) \ge \frac{1}{2}\|p - q\|_1^2$ relating Kullback–Leibler divergence to ℓ_1 distance) gives

$$D_\Psi(w_{t+1}\|w_t) \ge \tfrac{1}{2}\|\Delta_t\|_1^2, \qquad D_\Psi(w_t\|w_{t+1}) \ge \tfrac{1}{2}\|\Delta_t\|_1^2,$$

so the left-hand side is at least $\|\Delta_t\|_1^2$. Therefore

$$\|\Delta_t\|_1^2 \le \eta \, \|g_t\|_\infty \, \|\Delta_t\|_1,$$

which implies $\|w_{t+1} - w_t\|_1 = \|\Delta_t\|_1 \le \eta \|g_t\|_\infty$. □

Lemma 6 (Cumulative movement)**.** *For every integer $i \ge 1$,*

$$\|w_{t-i} - w_t\|_1 \le i \, \eta L.$$

Proof. By the triangle inequality and Lemma 5,

$$\|w_{t-i} - w_t\|_1 \le \sum_{j=1}^{i} \|w_{t-j} - w_{t-j+1}\|_1 \le i \, \eta L.$$

□

Lemma 7 (Action bounds relative to the comparator trajectory)**.** *For all t,*

$$u_t^{\mathcal{A}} \geq \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle - (1-\kappa)[b_t^{\pi(w_t)} - b_t^{\mathcal{A}}]_+, \tag{18}$$

$$u_t^{\mathcal{A}} \leq \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle. \tag{19}$$

Proof. Recall that

$$\widehat{u}_t = \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle, \qquad u_t^{\mathcal{A}} = \min\{\widehat{u}_t, b_t^{\mathcal{A}}\}.$$

Therefore,

$$u_t^{\mathcal{A}} = \widehat{u}_t - \left[\widehat{u}_t - b_t^{\mathcal{A}}\right]_+ = \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle - \left[\langle w_t, \widetilde{a}_t \rangle - (1-\kappa) b_t^{\mathcal{A}}\right]_+.$$

This identity gives the upper bound immediately, since the subtracted term is nonnegative.

For the lower bound, Lemma 3 applied to the comparator policy $\pi(w_t)$ yields

$$u_t^{\pi(w_t)} = \kappa b_t^{\pi(w_t)} + \langle w_t, \widetilde{a}_t \rangle \leq b_t^{\pi(w_t)}.$$

Hence

$$\langle w_t, \widetilde{a}_t \rangle \leq (1-\kappa) b_t^{\pi(w_t)}.$$

Subtracting $(1-\kappa)b_t^{\mathcal{A}}$ from both sides and applying monotonicity of $[\cdot]_+$ gives

$$\left[\langle w_t, \widetilde{a}_t \rangle - (1-\kappa) b_t^{\mathcal{A}}\right]_+ \leq (1-\kappa)\left[b_t^{\pi(w_t)} - b_t^{\mathcal{A}}\right]_+.$$

Substituting this bound into the expression above for $u_t^{\mathcal{A}}$ yields the stated lower bound. $\square$

Lemma 8 (State-gap recursion)**.** *Define the state gap*

$$\Gamma_t^w \triangleq b_t^{\pi(w)} - b_t^{\mathcal{A}}. \tag{20}$$

For any fixed $w \in \Delta_H$ and every $t \geq 1$ the following two-sided recursion holds:

$$\begin{aligned} \Gamma_{t+1}^w &= \Gamma_t^w - \left(u_t^{\pi(w)} - u_t^{\mathcal{A}}\right) \\ &\geq (1-\kappa)\Gamma_t^w + \langle w_t - w, \widetilde{a}_t \rangle - (1-\kappa)[\Gamma_t^{w_t}]_+, \end{aligned} \tag{21}$$

$$\Gamma_{t+1}^w \leq (1-\kappa)\Gamma_t^w + \langle w_t - w, \widetilde{a}_t \rangle. \tag{22}$$

where $[x]_+ := \max\{x, 0\}$.

Proof. From the state updates (10) (for $\pi(w)$) and (13) (for the algorithm) we have the exact identity

$$\Gamma_{t+1}^{w} = b_{t+1}^{\pi(w)} - b_{t+1}^{\mathcal{A}} = \Gamma_t^w - \big(u_t^{\pi(w)} - u_t^{\mathcal{A}}\big). \tag{23}$$

Consequently, bounding Γ_{t+1}^{w} reduces to bounding the action difference $u_t^{\pi(w)} - u_t^{\mathcal{A}}$.

Using the SDAC representation for the comparator action and the bounds of Lemma 7 for the algorithm action, we have

$$u_t^{\pi(w)} = \kappa b_t^{\pi(w)} + \langle w, \widetilde{a}_t \rangle,$$

and

$$\kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle - (1-\kappa)[\Gamma_t^{w_t}]_+ \le u_t^{\mathcal{A}} \le \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle.$$

Subtracting the upper bound on $u_t^{\mathcal{A}}$ from $u_t^{\pi(w)}$ yields the lower bound on $u_t^{\pi(w)} - u_t^{\mathcal{A}}$:

$$\begin{aligned} u_t^{\pi(w)} - u_t^{\mathcal{A}} &\ge \kappa b_t^{\pi(w)} + \langle w, \widetilde{a}_t \rangle \\ &\quad - \big(\kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle\big) \\ &= \kappa \Gamma_t^w - \langle w_t - w, \widetilde{a}_t \rangle. \end{aligned}$$

Inserting this into the identity (23) gives

$$\Gamma_{t+1}^{w} \le \Gamma_t^w - \big(\kappa \Gamma_t^w - \langle w_t - w, \widetilde{a}_t \rangle\big) = (1-\kappa)\Gamma_t^w + \langle w_t - w, \widetilde{a}_t \rangle,$$

which is (22).

Similarly, using the lower bound on $u_t^{\mathcal{A}}$ produces an upper bound on $u_t^{\pi(w)} - u_t^{\mathcal{A}}$:

$$\begin{aligned} u_t^{\pi(w)} - u_t^{\mathcal{A}} &\le \kappa b_t^{\pi(w)} + \langle w, \widetilde{a}_t \rangle \\ &\quad - \Big(\kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle - (1-\kappa)[\Gamma_t^{w_t}]_+\Big) \\ &= \kappa \Gamma_t^w - \langle w_t - w, \widetilde{a}_t \rangle + (1-\kappa)[\Gamma_t^{w_t}]_+. \end{aligned}$$

Substituting this bound into the identity (23) yields

$$\Gamma_{t+1}^{w} \ge \Gamma_t^w - \big(\kappa \Gamma_t^w - \langle w_t - w, \widetilde{a}_t \rangle + (1-\kappa)[\Gamma_t^{w_t}]_+\big) = (1-\kappa)\Gamma_t^w + \langle w_t - w, \widetilde{a}_t \rangle - (1-\kappa)[\Gamma_t^{w_t}]_+,$$

which is (21) and completes the proof. $\square$

Lemma 9 (State deviation bound)**.** *For every $t \geq 1$, let the state gap be defined by* (20)*. Then*

$$-\frac{\eta L}{\kappa^3} \leq \Gamma_t^{w_t} = b_t^{\pi(w_t)} - b_t^{\mathcal{A}} \leq \frac{\eta L}{\kappa^2}.$$

Proof. From (22), for any fixed comparator $w \in \Delta_H$ and every $s \geq 1$,

$$\Gamma_{s+1}^w \leq (1-\kappa)\Gamma_s^w + \langle w_s - w, \widetilde{a}_s \rangle.$$

Applying the recursion repeatedly up to time t yields

$$\begin{aligned}
\Gamma_t^w &\leq (1-\kappa)\Gamma_{t-1}^w + \langle w_{t-1} - w, \widetilde{a}_{t-1} \rangle \\
&\leq (1-\kappa)^2\Gamma_{t-2}^w + (1-\kappa)\langle w_{t-2} - w, \widetilde{a}_{t-2} \rangle + \langle w_{t-1} - w, \widetilde{a}_{t-1} \rangle \\
&\ \ \vdots \\
&\leq (1-\kappa)^{t-1}\Gamma_1^w + \sum_{s=1}^{t-1}(1-\kappa)^{t-1-s}\langle w_s - w, \widetilde{a}_s \rangle.
\end{aligned}$$

Now fix $w = w_t$. Since $b_1^{\pi(w_t)} = b_1^{\mathcal{A}} = 0$, we have $\Gamma_1^{w_t} = 0$, and therefore

$$\Gamma_t^{w_t} \leq \sum_{s=1}^{t-1}(1-\kappa)^{t-1-s}\langle w_s - w_t, \widetilde{a}_s \rangle.$$

Note that $\|\widetilde{a}_s\|_\infty = \max_{i \leq H}(1-\kappa)^i a_{s-i} \leq 1-\kappa$ since $a_\tau \in [0,1]$ and $i \geq 1$. Hence,

$$\begin{aligned}
\Gamma_t^{w_t} &\leq \sum_{s=1}^{t-1}(1-\kappa)^{t-1-s}\big|\langle w_s - w_t, \widetilde{a}_s \rangle\big| \\
&\leq \sum_{s=1}^{t-1}(1-\kappa)^{t-1-s}\|w_s - w_t\|_1\|\widetilde{a}_s\|_\infty \\
&\leq (1-\kappa)\eta L\sum_{s=1}^{t-1}(1-\kappa)^{t-1-s}(t-s) \\
&= (1-\kappa)\eta L\sum_{j=1}^{t-1} j(1-\kappa)^{j-1} \leq \eta L\sum_{j=1}^{\infty} j(1-\kappa)^{j-1} = \frac{\eta L}{\kappa^2}.
\end{aligned}$$

Where we used the geometric-series identity

$$\sum_{j=1}^{\infty} j(1-\kappa)^{j-1} = \frac{1}{\kappa^2}.$$

Similarly, for the lower bound, unrolling (21) with the same fixed comparator $w = w_t$ yields

$$\Gamma_t^{w_t} \geq \sum_{s=1}^{t-1} (1-\kappa)^{t-1-s} \langle w_s - w_t, \widetilde{a}_s \rangle - \sum_{s=1}^{t-1} (1-\kappa)^{t-s} [\Gamma_s^{w_s}]_+.$$

The upper bound just proved applies to every earlier time s, so $[\Gamma_s^{w_s}]_+ \leq \eta L/\kappa^2$. Also, by Hölder's inequality and Lemma 6,

$$\begin{aligned}
\Gamma_t^{w_t} &\geq -\eta L \sum_{s=1}^{t-1} (1-\kappa)^{t-1-s} (t-s) \\
&\quad - \frac{\eta L}{\kappa^2} \sum_{s=1}^{t-1} (1-\kappa)^{t-s} \\
&= -\eta L \sum_{j=1}^{t-1} j(1-\kappa)^{j-1} - \frac{\eta L}{\kappa^2} \sum_{j=1}^{t-1} (1-\kappa)^j.
\end{aligned}$$

Extending both sums to infinity gives

$$\Gamma_t^{w_t} \geq -\eta L \sum_{j=1}^{\infty} j(1-\kappa)^{j-1} - \frac{\eta L}{\kappa^2} \sum_{j=1}^{\infty} (1-\kappa)^j.$$

Using the geometric-series identities

$$\sum_{j=1}^{\infty} j(1-\kappa)^{j-1} = \frac{1}{\kappa^2}, \qquad \sum_{j=1}^{\infty} (1-\kappa)^j = \frac{1-\kappa}{\kappa},$$

we obtain

$$\Gamma_t^{w_t} \geq -\frac{\eta L}{\kappa^2} - \frac{\eta L(1-\kappa)}{\kappa^3} \geq -\frac{\eta L}{\kappa^3}.$$

Hence, the claim. □

Lemma 10 (Action-state gap)**.** *Let the state gap be defined by* (20)*. Then for all* t*,*

$$-\frac{\eta L}{\kappa^2} \leq \kappa \Gamma_t^{w_t} \leq u_t^{\pi(w_t)} - u_t^{\mathcal{A}} \leq \kappa \Gamma_t^{w_t} + (1-\kappa)[\Gamma_t^{w_t}]_+ \leq \frac{\eta L}{\kappa^2}.$$

Proof. **Upper bound.** From Lemma 7 (lower bound (18)),

$$u_t^{\mathcal{A}} \geq \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle - (1-\kappa)\, [\Gamma_t^{w_t}]_+.$$

Subtracting from $u_t^{\pi(w_t)} = \kappa b_t^{\pi(w_t)} + \langle w_t, \widetilde{a}_t \rangle$ (by (9)),

$$u_t^{\pi(w_t)} - u_t^{\mathcal{A}} \leq \kappa(b_t^{\pi(w_t)} - b_t^{\mathcal{A}}) + (1-\kappa)\,[\Gamma_t^{w_t}]_+ = \kappa\Gamma_t^{w_t} + (1-\kappa)\,[\Gamma_t^{w_t}]_+.$$

Here we used (20).

Lower bound. From Lemma 7 (upper bound (19)),

$$u_t^{\mathcal{A}} \leq \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle,$$

so

$$u_t^{\mathcal{A}} - u_t^{\pi(w_t)} \leq \kappa b_t^{\mathcal{A}} + \langle w_t, \widetilde{a}_t \rangle - (\kappa b_t^{\pi(w_t)} + \langle w_t, \widetilde{a}_t \rangle) = \kappa(b_t^{\mathcal{A}} - b_t^{\pi(w_t)}) = -\kappa\Gamma_t^{w_t}.$$

Equivalently,

$$u_t^{\pi(w_t)} - u_t^{\mathcal{A}} \geq \kappa\Gamma_t^{w_t}.$$

Finally, by Lemma 9,

$$\Gamma_t^{w_t} \geq -\frac{\eta L}{\kappa^3} \quad \Longrightarrow \quad \kappa\Gamma_t^{w_t} \geq -\frac{\eta L}{\kappa^2},$$

and also

$$[\Gamma_t^{w_t}]_+ \leq \frac{\eta L}{\kappa^2}.$$

Hence

$$\kappa\Gamma_t^{w_t} + (1-\kappa)[\Gamma_t^{w_t}]_+ \leq \frac{\eta L}{\kappa^2},$$

which gives the rightmost inequality and completes the proof. □

Lemma 11 (Stability term). *For all horizons T,*

$$\sum_{t=1}^{T} \left(c_t(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - c_t\left(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\right)\right) \leq \frac{T\eta L^2}{\kappa^2},$$

where $L = L_u + \frac{L_b}{\kappa}$.

Proof. By Lipschitz continuity of c_t (see equation (3)),

$$c_t\big(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) \leq L_u\big|u_t^{\mathcal{A}} - u_t^{\pi(w_t)}\big| + L_b\big|b_t^{\mathcal{A}} - b_t^{\pi(w_t)}\big|.$$

From Lemma 10,

$$|u_t^{\mathcal{A}} - u_t^{\pi(w_t)}| \leq \frac{\eta L}{\kappa^2},$$

Also, by Lemma 9 and (20),

$$|b_t^{\mathcal{A}} - b_t^{\pi(w_t)}| = |\Gamma_t^{w_t}| \leq \frac{\eta L}{\kappa^3}.$$

Therefore, for each $t \geq 1$,

$$c_t\big(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}\big) - c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) \leq L_u \frac{\eta L}{\kappa^2} + L_b \frac{\eta L}{\kappa^3}.$$

Summing over $t = 1, 2, \ldots, T$ gives

$$\sum_{t=1}^{T} \Big(c_t(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) \Big) \leq T \frac{\eta L}{\kappa^2} \left(L_u + \frac{L_b}{\kappa} \right) = \frac{T \eta L^2}{\kappa^2},$$

where we used $L = L_u + \frac{L_b}{\kappa}$ in the final step. □

Proof of Theorem 1. Starting from (5) we have the following decomposition of the regret:

$$\begin{aligned}
\mathrm{Regret}(T) &= \sum_{t=1}^{T} c_t(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \min_{w \in \Delta_H} \sum_{t=1}^{T} c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big) \\
&= \sum_{t=1}^{T} c_t(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \sum_{t=1}^{T} c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) + \sum_{t=1}^{T} c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) - \min_{w \in \Delta_H} \sum_{t=1}^{T} c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big).
\end{aligned}$$

We call the first sum the stability term, and the second sum the OMD term. We will bound each term separately.

By standard entropic OMD (see for example [4]) analysis on the surrogate losses $\ell_t(w)$ defined in (14), which are convex and L-Lipschitz by Lemma 4, we obtain

$$\sum_{t=1}^{T} \ell_t(w_t) - \min_{w \in \Delta_H} \sum_{t=1}^{T} \ell_t(w) \leq \frac{\log H}{\eta} + \frac{\eta L^2 T}{2}.$$

Using (14), we substitute $\ell_t(w) = c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big)$, so that

$$\sum_{t=1}^{T} \ell_t(w_t) = \sum_{t=1}^{T} c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) \quad \text{and} \quad \min_{w \in \Delta_H} \sum_{t=1}^{T} \ell_t(w) = \min_{w \in \Delta_H} \sum_{t=1}^{T} c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big).$$

Thus this term becomes

$$\sum_{t=1}^{T} c_t\big(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}\big) - \min_{w \in \Delta_H} \sum_{t=1}^{T} c_t\big(b_t^{\pi(w)}, u_t^{\pi(w)}\big) \leq \frac{\log H}{\eta} + \frac{\eta L^2 T}{2}.$$

From Lemma 11, the stability term satisfies

$$\sum_{t=1}^{T} c_t(b_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \sum_{t=1}^{T} c_t(b_t^{\pi(w_t)}, u_t^{\pi(w_t)}) \leq \frac{\eta L^2 T}{\kappa^2}.$$

Adding the OMD term and the stability term yields the combined bound

$$\mathrm{Regret}(T) \;\leq\; \frac{\log H}{\eta} \;+\; \frac{\eta L^2 T}{2} \;+\; \frac{\eta L^2 T}{\kappa^2}.$$

Rearranging by factoring out the η term gives

$$\mathrm{Regret}(T) \;\leq\; \frac{\log H}{\eta} \;+\; \eta L^2 T\Big(\tfrac{1}{2} + \tfrac{1}{\kappa^2}\Big).$$

To optimize over $\eta > 0$, we apply the inequality $A/\eta + B\eta \leq 2\sqrt{AB}$ with $A = \log H$ and $B = L^2 T(\frac{1}{2} + \frac{1}{\kappa^2})$. This yields the optimal step size

$$\eta^* = \sqrt{\frac{\log H}{L^2 T\big(\frac{1}{2} + \frac{1}{\kappa^2}\big)}}$$

and the resulting regret bound

$$\mathrm{Regret}(T) \;\leq\; 2L\sqrt{\Big(\tfrac{1}{2} + \tfrac{1}{\kappa^2}\Big) T \log H}.$$

□

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